Memory

0x0 - 0x37

0x4520 - 0x4527

Displaying Memory Locations from 0x4520 to 0x4527

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4520	05
0x4521	04
0x4522	14
0x4523	00
0x4524	14
0x4525	03
0x4526	06
0x4527	02

```
1  LDA 4520H
2  MOV E,A
3  MVI D,00H
4  LDA 4521H
5  MOV C,A
6  LXI H,0000H
7  MOV A,C
8  ORA A
9  JZ STORE
10 LOOP: DAD D
11 DCR C
12 JNZ LOOP
13 STORE: SHLD 4522H
14 LDA 4525H
15 MOV B,A
16 ORA A
17 JZ STOP
18 LDA 4524H
19 MVI C,00H
20 NEXT: CMP B
21 JC DONE
22 SUB B
23 INR C
24 JMP NEXT
25 DONE: STA 4527H
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 20 45	LDA 4520H
2	0x3	5F	MOV E,A
3	0x4	16 00	MVI D,00H
4	0x6	3A 21 45	LDA 4521H
5	0x9	4F	MOV C,A
6	0xA	21 00 00	LXI H,0000H
7	0xD	79	MOV A,C
8	0xE	B7	ORA A
9	0xF	CA 17 00	JZ STORE
10	0x12	19	LOOP: DAD D
11	0x13	0D	DCR C
12	0x14	C2 12 00	JNZ LOOP
13	0x17	22 22 45	STORE: SHLD 4522H

Registers

A/PSW	0x 00 44
BC	0x 00 00
DE	0x 00 00
HL	0x 00 00
SP	0x FF FF
PC	0x 00 38

Flags

S	☐
Z	☑
AC	☐
P	☑

```
4  LDA 4521H
5  MOV C,A
6  LXI H,0000H
7  MOV A,C
8  ORA A
9  JZ STORE
10 LOOP: DAD D
11 DCR C
12 JNZ LOOP
13 STORE: SHLD 4522H
14 LDA 4525H
15 MOV B,A
16 ORA A
17 JZ STOP
18 LDA 4524H
19 MVI C,00H
20 NEXT: CMP B
21 JC DONE
22 SUB B
23 INR C
24 JMP NEXT
25 DONE: STA 4527H
26 MOV A,C
27 STA 4526H
28 STOP: HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 20 45	LDA 4520H
2	0x3	5F	MOV E,A
3	0x4	16 00	MVI D,00H
4	0x6	3A 21 45	LDA 4521H
5	0x9	4F	MOV C,A
6	0xA	21 00 00	LXI H,0000H
7	0xD	79	MOV A,C
8	0xE	B7	ORA A
9	0xF	CA 17 00	JZ STORE
10	0x12	19	LOOP: DAD D
11	0x13	0D	DCR C
12	0x14	C2 12 00	JNZ LOOP

Given:

multiplicand = 05H = 5

Multiplier = 04H = 4

Calculation:

$5 \times 4 = 20$

$20_{10} = 14H$

The 16-bit product is 0014H

Division:

Given:

Dividend = 14H = 20

Divisor = 03H = 3

$20 = (3 \times 6) + 2$

∴ Quotient = 6 = 06H

Remainder

20 - 3 17

17 - 3 14

14 - 3 11

11 - 3 8

8 - 3 5

5 - 3 2

The Subtraction is performed